

Mayo Clinic Study of Aging, part 2

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21 May 2026

1 Assessing amyloid

A key question of the paper was the effect of amyloid levels on the progression to dementia. Is this expensive test worth doing? Key issues:

- Age is the most important covariate
- Presence of one or more $\epsilon 4$ alleles in the APOE gene are known to increase the risk of dementia. Some of that may be via increased amyloid deposition.
- There is a known sex*APOE interaction, with respect to the effect.
- About 40% of the subjects have amyloid. There was prior work using categorical amyloid.
- The PiB tracer did not become available until partway into the MCSA, those who have amyloid don't all start imaging at visit 0.

2 Hazard ratios

```
library(survival)
library(mshmm)
```

Loading required package: Matrix

```
library(splines)
table2 <- function(...) table(..., useNA= "ifany")
options(show.significant.stars = FALSE)

dx2 <- factor(c(1,2,2,3,2,3,4)[as.numeric(dementia$dx)], 1:4,
              c("alive", "ND", "dementia", "death"))
table2(dementia$dx, dx2)
```

	dx2			
	alive	ND	dementia	death
alive	2680	0	0	0

CU	0	17264	0	0
MCI	0	2357	0	0
dementia	0	0	395	0
nondem	0	1074	0	0
off-dem	0	0	430	0
death	0	0	0	1471

```

if (packageVersion("survival") >= "3.8-7") {
  print(survcheck(Surv(age, dx2) ~1, dementia, id= ptnum)$transitions)

  # create the simpler endpoint of not demented (ND), ever dementia, death
  temp <- unlist(tapply(as.numeric(dx2), dementia$ptnum, function(z)
    ifelse(z==1, 1, cummax(z))))
  dx2 <- factor(temp, 1:4, levels(dx2))
  survcheck(Surv(age, dx2) ~1, dementia, id= ptnum)$transitions

  dem2 <- timeline2counting(Surv(age, dx2) ~ ., data=dementia, id=ptnum)
  subset(dem2, ptnum <3, -c(p.tau181, memory, global, hpvol))

  # Add in amyloid group of low/med/high/missing
  temp <- with(dem2, ifelse(is.na(centiloid), 4,
    cut(centiloid, c(-50, 25, 68, 500))))
  dem2$agrp <- factor(temp, 1:4, c("low", "mod", "high", "miss"))
  table(dem2$agrp)

  # add in time since enrollment as 0-1, 1-4, 4+ years, with 4+ as reference
  indx <- match(dem2$ptnum, dem2$ptnum)
  dem2$year1 <- dem2$age1 - dem2$age1[indx]
  dem2$year2 <- dem2$age2 - dem2$age1[indx]
  dem3 <- survSplit(Surv(year1, year2, dx2) ~ ., data=dem2,
    cut=c(1,4), episode= "dxyr")

  dem3$dxyr <- factor(dem3$dxyr, c(3,1,2), c("4+", "0-1", "1-4"))
  indx <- match(dem3$ptnum, dem3$ptnum)
  dem3$age1 <- dem3$year1 + dem3$age1[indx]
  dem3$age2 <- dem3$year2 + dem3$age1[indx]

  # education per 4 years
  dem3$edu4 <- dem3$educ/4
  dem3 <- subset(dem3, !is.na(edu4))
} else dem3 <- readRDS("data/dem3.rds")

```

	to	ND	dementia	death	(censored)
from					
ND		0	662	971	2713
dementia		29	0	500	132

```

death      0      0      0      0

survcheck(Surv(age1, age2, dx2) ~ 1, data=dem3, id=ptnum, istate=istate)

```

```

Call:
survcheck(formula = Surv(age1, age2, dx2) ~ 1, data = dem3, id = ptnum,
  istate = istate)

```

Unique identifiers	Observations	Transitions
4313	28334	2124

Transitions table:

	to			
from	dementia	death	(censored)	
ND	653	957	2703	
dementia	0	514	138	
death	0	0	0	

Number of subjects with 0, 1, ... transitions to each state:

	count			
state	0	1	2	
dementia	3660	653	0	
death	2842	1471	0	
(any)	2703	1096	514	

```

fit1 <- coxph(list(Surv(age1, age2, dx2) ~ male + any.e4 + agrp + cmc + edu4,
  1:2 ~ male*(any.e4 + agrp),
  1:3 ~ dxyr), data=dem3, id=ptnum, istate= istate)
print(fit1, digits=2)

```

```

Call:
coxph(formula = list(Surv(age1, age2, dx2) ~ male + any.e4 +
  agrp + cmc + edu4, 1:2 ~ male * (any.e4 + agrp), 1:3 ~ dxyr),
  data = dem3, id = ptnum, istate = istate)

```

1:2	coef	exp(coef)	robust se	z	p
male	0.990	2.692	0.332	3.0	0.003
any.e4	0.778	2.177	0.113	6.9	5e-12
agrpmo	0.980	2.665	0.418	2.3	0.019
agrphigh	1.789	5.981	0.332	5.4	7e-08
agrpmis	1.066	2.903	0.297	3.6	3e-04
cmc	0.109	1.115	0.028	3.9	1e-04
edu4	-0.057	0.944	0.051	-1.1	0.265
male:any.e4	-0.394	0.674	0.163	-2.4	0.015
male:agrpmo	-0.930	0.394	0.507	-1.8	0.067

```
male:agrphigh -0.890      0.411      0.404 -2.2 0.028
male:agrpmis -0.846      0.429      0.346 -2.4 0.014
```

```
1:3      coef exp(coef) robust se      z      p
male      0.318      1.375      0.069  4.6 4e-06
any.e4    -0.036      0.965      0.078 -0.5  0.6
agrpmo     0.134      1.143      0.185  0.7  0.5
agrphigh  -0.053      0.948      0.188 -0.3  0.8
agrpmis    0.515      1.673      0.110  4.7 3e-06
cmc        0.298      1.347      0.025 12.1 <2e-16
edu4       -0.054      0.948      0.045 -1.2  0.2
dxyr0-1    -0.997      0.369      0.183 -5.4 5e-08
dxyr1-4    -0.308      0.735      0.085 -3.6 3e-04
```

```
2:3      coef exp(coef) robust se      z      p
male      0.08085      1.08421      0.08810  0.9 0.36
any.e4     0.02059      1.02080      0.08819  0.2 0.82
agrpmo    -0.01603      0.98409      0.24563 -0.1 0.95
agrphigh  -0.00063      0.99937      0.20203  0.0 1.00
agrpmis    0.09603      1.10080      0.17707  0.5 0.59
cmc        0.13909      1.14922      0.03285  4.2 2e-05
edu4       0.09083      1.09508      0.05354  1.7 0.09
```

States: 1= ND, 2= dementia, 3= death

Likelihood ratio test=452 on 27 df, p=<2e-16
n= 28324, unique id= 4313, number of events= 2123
(10 observations deleted due to missingness)

```
print(coef(fit1, matrix=TRUE), digits=2)
```

```
      1:2      1:3      2:3
male      0.990  0.318  0.08085
any.e4     0.778 -0.036  0.02059
agrpmo     0.980  0.134 -0.01603
agrphigh   1.789 -0.053 -0.00063
agrpmis    1.066  0.515  0.09603
cmc        0.109  0.298  0.13909
edu4       -0.057 -0.054  0.09083
dxyr0-1     0.000 -0.997  0.00000
dxyr1-4     0.000 -0.308  0.00000
male:any.e4 -0.394  0.000  0.00000
male:agrpmo -0.930  0.000  0.00000
male:agrphigh -0.890  0.000  0.00000
```

```

male:agrpmiss -0.846 0.000 0.00000
attr(,"states")
[1] "ND"      "dementia" "death"

fit1$cmap

          1:2 1:3 2:3
male          1 12 21
any.e4         2 13 22
agrpmod        3 14 23
agrphigh       4 15 24
agrpmiss       5 16 25
cmc            6 17 26
edu4           7 18 27
dxyr0-1        0 19  0
dxyr1-4        0 20  0
male:any.e4     8  0  0
male:agrpmod    9  0  0
male:agrphigh 10  0  0
male:agrpmiss 11  0  0

```

Because of the interactions, simple coefficients for the key variables (male, e4, amyloid) that affect the ND:dementia transition are difficult to interpret. A cleaner approach is to compute and contrast predicted values, which have the further advantage that they are invariant to how dummy variables are coded. The contrast.R file contains a simple function for computing these predicted values and their standard errors, and marginal.Rnw contains a longer discussion of the concepts. Because education and CMC are additive in the model, we can choose any value for them in the dummy data, the contrast is invariant.

```

source("contrast.R")
dummy <- expand.grid(any.e4= 0:1, agrp=c("low", "mod", "high", "miss"),
                    male=0:1, dxyr="4+", edu4=3, cmc=1)

# get overall population counts for the categories
temp <- with(dem3, table(any.e4, agrp, male))
wt <- as.vector(temp/ sum(temp))

# The contrast function isn't yet quite smart enough for multistate.
# But we only need to use it for the ND:dementia transition, so so a
# sub fit.
fit1a <- coxph(Surv(age1, age2, dx2=="dementia") ~ male*(any.e4 + agrp) +
              cmc+ edu4, data= dem3, ties="breslow",
              subset=(istate=="ND"))
round(rbind(coef(fit1a), coef(fit1)[1:11]), 4)

      male any.e4 agrpmod agrphigh agrpmiss      cmc      edu4

```

```

[1,] 0.9904 0.7778 0.9803 1.7887 1.0656 0.1089 -0.0573
[2,] 0.9904 0.7778 0.9803 1.7887 1.0656 0.1089 -0.0573
      male:any.e4 male:agrpmod male:agrphigh male:agrpmiss
[1,]      -0.3939      -0.9303      -0.8897      -0.8456
[2,]      -0.3939      -0.9303      -0.8897      -0.8456

```

```

cmat <- contrast(fit1a, dummy, weight=wt)
yy1 <- cmat[,1] + outer(cmat[,2], c(0, -1.96, 1.96), '*')

```

```

# the other, non-interaction coefs
temp <- summary(fit1)$conf.int
yy2 <- log(temp[-c(1:5, 8:11), c(1,3,4)])

```

Now create the plots. First create labels for the columns

```

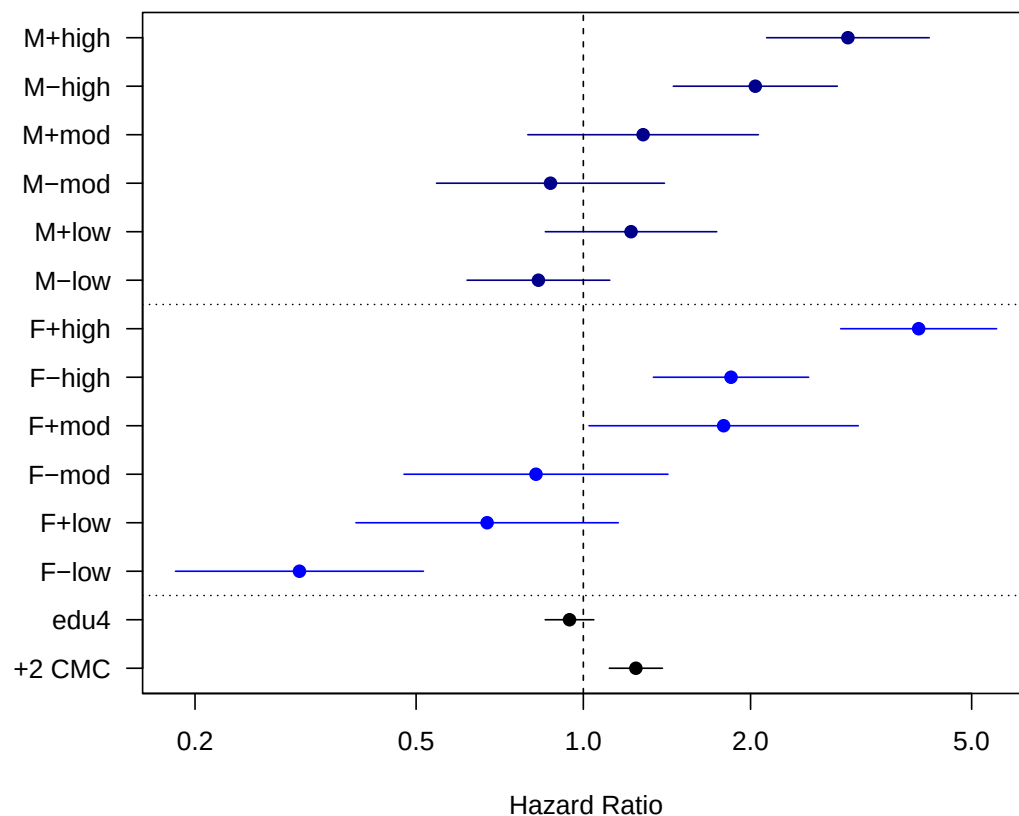
p1 <- outer(c("F", "M"), c( "-low", "+low", "-mod", "+mod", "-high", "+high",
                             "-miss", "+miss"), paste0)
p1 <- c( "+2 CMC", "edu4", t(p1))
jj <- grepl("cmc", rownames(yy2))
yy2[jj,] <- yy2[jj,] *2

opar <- par(mar=c(5,6,1,1))

z1 <- exp(rbind( yy2[1:2,], yy1[c(1:6, 9:14),]))
matplot(z1, 1:14, type='n', log='x', xlab="Hazard Ratio", ylab="",
        yaxt='n')
pp1 <- p1[c(1:8, 11:16)]
axis(side=2, 1:14, pp1, las=2, cex.axis=1)

icol <- rep(c("black", "blue", "darkblue"), c(2,6,6))
points(z1[,1], 1:14, pch=19, col=icol)
segments(z1[,2], 1:14, z1[,3], 1:14, col=icol)
abline(v=1, lty=2)
abline(h=c(2.5, 8.5), lty=3)

```



```
par(opar)
```

The above plot is nearly identical to the top panel of figure 2 in Jack 2022. The dementia data in the package is not the data set used in the paper, but is heavily based on it, e.g., the same pattern of visits (fraction with 1, 2, etc.), but fewer subjects, no dates, artificial subject identifiers, random noise added to all continuous variables and to all between visit time intervals, times rounded to .1 year, The final data set is intended to be a close enough simulant for the exploration of statistical issues. It should not be used for medical decision making.

Below is a plot that includes all the hazard ratios.

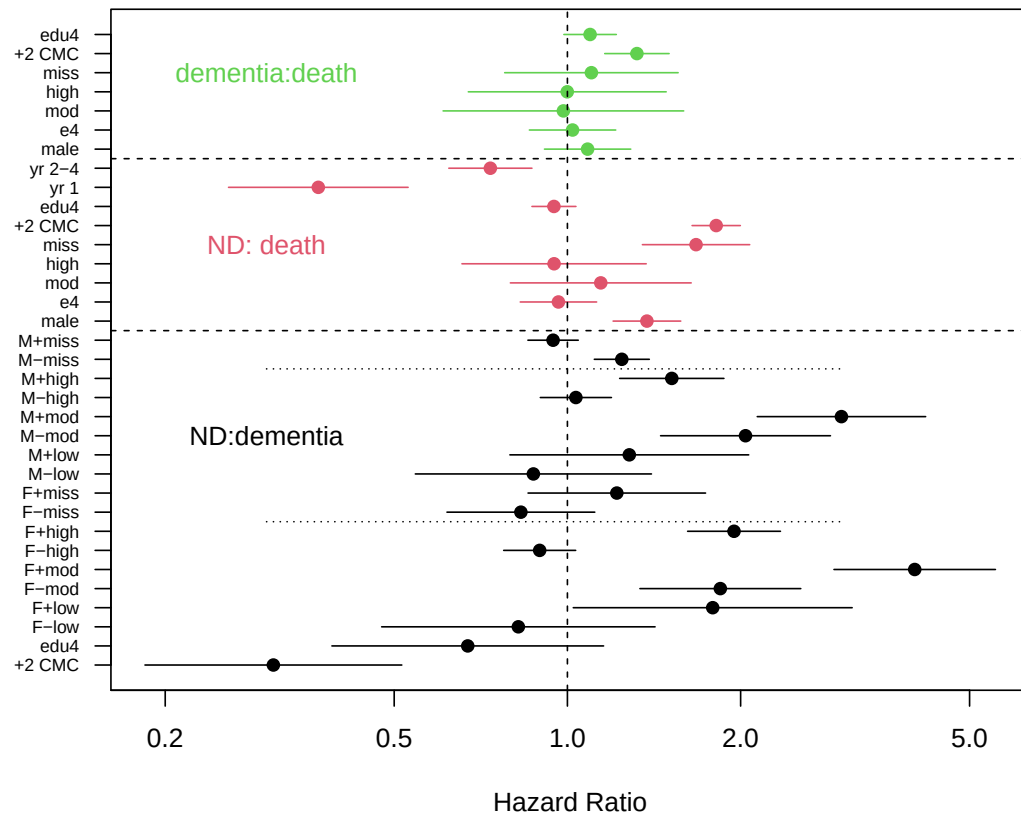
```
z2 <- exp(rbind(yy1, yy2))
p2 <- c("male", "e4", "mod", "high", "miss", "+2 CMC", "edu4")
p2 <- c(p2, "yr 1", "yr 2-4", p2)

opar <- par(mar=c(5,5,1,1))
matplot(z2, 1:34, type='n', log='x', xlab="Hazard Ratio", ylab="",
        yaxt='n')
```

```

axis(side=2, 1:34, c(p1,p2), las=2, cex.axis=.7)
points(z2[,1], 1:34, pch=19, col=rep(1:3, c(18, 9,7)))
segments(z2[,2], 1:34, z2[,3],1:34, col=rep(1:3, c(18, 9,7)))
abline(v=1, lty=2)
segments(c(.3, .3), c(8.5, 16.5), c(3,3), c(8.5, 16.5), lty=3)
abline(h=c(18.5, 27.5), lty=2)
text(c(.3, .3, .3), c(13, 23, 32), col=1:3,
      c("ND:dementia", "ND: death", "dementia:death"))

```



```

par(opar)

```

3 Event rates

Overall hazard rates can be computed using a simple poisson regression.

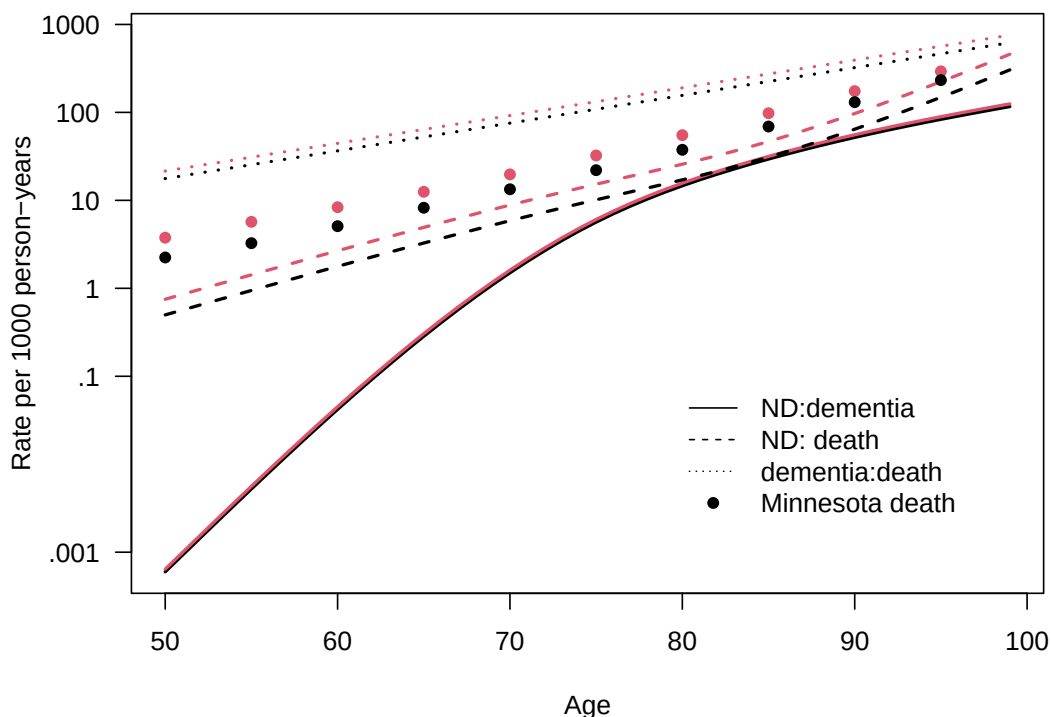
```

tdata <- survSplit(Surv(age1, age2, dx2) ~ ., data=dem3, cut=51:99)

delta <- tdata$age2 - tdata$age1 # interval length
rate1 <- glm(I(dx2=="dementia") ~ male + ns(age1,3) + offset(log(delta)),
             data=tdata, poisson, subset= (istate=="ND"))
rate2 <- glm(I(dx2=="death") ~ male + ns(age1,3) + dxyr + offset(log(delta)),
             data=tdata, poisson, subset= (istate=="ND"))
rate3 <- glm(I(dx2=="death") ~ male + age1 + offset(log(delta)),
             data=tdata, poisson, subset= (istate=="dementia"))
dummy <- expand.grid(age1=50:99, male=0:1, dxyr = "4+", delta=1000)
yhat1 <- matrix(predict(rate1, newdata=dummy, type="response"), ncol=2)
yhat2 <- matrix(predict(rate2, newdata=dummy, type="response"), ncol=2)
yhat3 <- matrix(predict(rate3, newdata=dummy, type="response"), ncol=2)

matplot(50:99, cbind(yhat1, yhat2, yhat3), log='y', type='l', lwd=2,
        lty=c(1,1,2,2,3,3), col=1:2, yaxt='n',
        xlab= "Age", ylab="Rate per 1000 person-years")
axis(2, c(.001, .1, 1, 10, 100, 1000), c(".001", ".1", "1", "10", "100", "1000"),
     las=2)
legend(80, .1, c("ND:dementia", "ND: death", "dementia:death",
                 "Minnesota death"), lty=c(1:3,0), pch=c(NA,NA,NA,16), bty='n')
xx <- seq(50,95, 5)
matpoints(xx, survexp.mn[xx+1,, "2010"]*365250, col=2:1, pch=16)

```



The dementia rate for males is just a smidge higher than females, 7%, (10% in the real data, neither is statistically significant). The death rates for males are estimated at 1.5 and 1.2 times female rates in the non-demented and demented, respectively.

4 Absolute risk

For survival curves, my stance for several decades had been that “survival curves for a time-dependent covariate are an oxymoron”. My standard response was to use landmarking, i.e., use a time-fixed covariate at the origin of the curve. A realization when doing this analysis was that when using age as the time scale, this approach simply doesn’t work. Consider the curve for a 65 year old with low amyloid, as in figure 3 of the paper. The median follow-up time in the data is 7.6 years. This means that most of the subjects contributing data to the 85-90 portion of that curve, for example, were recruited at ages of 79–85. The amyloid at study entry for these is an amyloid at 79 to 85. A naive curve using “baseline amyloid” of 25 centiloids corresponds to a population of subjects whose amyloid can start to progress, but each is regularly replaced by a rejuvenated copy whose amyloid has not. This curve does not represent anybody.

How does one create a predicted survival curve for a 65 year old e4- male whose current

amyloid level is 25 centiloids, from a model that used amyloid as a time-dependent covariate. This requires that we first predict the progression of amyloid over time, and then use that in prediction of survival. The 2022 paper used an ad hoc version of this based on our long experience with using US life tables.

The life table is a large array containing death rates by age, sex, and calendar year. To create the predicted survival to age 80 for a 48.2 year old female starting on 6-6-2000, fetch the appropriate 1 day of hazard from the table, increment age and calendar year by 1 day, fetch the appropriate value, increment, ... until we reach the ending age. Accumulate this to create a cumulated hazard $\Lambda(t)$, the predicted survival is $\exp(-\Lambda(t))$. The `pyears` and `survexp` routines of the `survival` package implement exactly this algorithm.

For the dementia data, first use the fitted multistate model to create a death rate table by integer age, amyloid (4 levels) and CMC (8 levels); paralleling the structure of US rate tables. We then used Poisson models to compute predicted amyloid state occupancy over time for a given starting age and state, and likewise for CMC state. Whereas age and sex for subjects in the usual `survexp` computation procede forward in lockstep, here each forward step of one year implies an updated distribution over states, and a weighted average of the death rates was added to the cumulative hazard. Only later did we realize that this entire bookkeeping process mimics a multistate Aalen-Johansen estimate, and could be approach from that direction.

Yet to be done: compute the predicted curves using this method.

Big questions:

- One example + one simulation $\neq$ proof
- Why 3 bilirubin groups? Why not 6, 10, 100?
- What about multiple TD covariates?
- How does this compare to joint models?
 - joint model is true
 - frailty assumption false
 - linearity assumption false
- Interval censored models?